Shaik. Basheer
192465049

SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: **B.E / B.Tech**
Course Code: **MLA02**
Course Name: **FUNDAMENTALS OF MACHINE LEARNING**
Duration: **1:30 Hr**
Date : 31/10/2026

Max Marks: **50 Marks**

ASSIGNMENT – I

UNIT I – INTRODUCTION AND MATHEMATICAL FOUNDATIONS

Topics: *Design of a learning system, types of machine learning, probability theory and distributions, Bayes decision theory, information theory*

Course Outcome

On successful completion of this assignment, students will be able to:

CO	Course Outcome Statement
CO1	Analyse machine learning fundamentals and mathematical foundations, including probability theory, Bayes decision theory, and information theory.

Answer all the Tasks

Task 1. A retail bank plans five machine-learning systems. For each system, analyse the problem and state (i) the learning paradigm (supervised / unsupervised / semi-supervised / reinforcement), (ii) the learning task (classification, regression, clustering, control, etc.) and (iii) the experience E and performance measure P in Mitchell's (T, P, E) formulation:

(a) Predict whether a loan applicant will default, using ten years of labelled repayment records.

(b) Discover behavioural customer segments among 2 million customers for which no labels exist.

(c) Estimate the next-month credit-card spend (in ₹) of each customer.

(d) Learn an ATM cash-refill policy that maximises cash availability minus transport cost through repeated trial and feedback.

(e) Only 2% of 500,000 transaction records are labelled fraud/genuine; the rest are unlabelled. Build a fraud detector that exploits both kinds of data.

Finally, justify which of the five systems is most harmed by label noise.

(5) (CO1) (BL4)

Task 2. A screening test for a disease with prevalence 2% has sensitivity 95% and specificity 90%. Calculate the probability that a person who tests positive actually has the disease, and analyse why the answer is much lower than the sensitivity.

(2) (CO1) (BL4)

Task 3. The joint probability mass function of a discrete random vector (X, Y) is given below.

X \ Y	Y = 0	Y = 1
X = 0	0.10	0.20
X = 1	0.15	0.25

X \ Y	Y = 0	Y = 1
X = 2	0.20	0.10

Analyse the distribution by (a) finding the marginals of X and Y. (b) computing E[X], Var(X) and E[Y]. (c) computing $P(X = 2 \mid Y = 1)$. (d) computing Cov(X, Y) and the correlation coefficient, and (e) deciding whether X and Y are independent, with a justification.

(6) (CO1) (BL4)

Task 4. A temperature sensor output is modelled as $X \sim N(\mu = 50, \sigma^2 = 16)$.

(a) Calculate $P(X > 58)$ and $P(46 < X < 54)$.

(b) A second sensor gives the readings 12, 15, 11, 14, 13. Derive the maximum-likelihood estimates of its mean and variance and compare the ML variance with the unbiased estimate, stating which is biased and why. (Use $\Phi(1) = 0.8413$, $\Phi(2) = 0.9772$.)

(4) (CO1) (BL4)

Task 5. A source emits four symbols with probabilities (1/2, 1/4, 1/8, 1/8). Calculate its entropy, compare it with the entropy of a uniform source over four symbols, and analyse the redundancy. Also compute the average length of the code {0, 10, 110, 111} and comment on its optimality.

(3) (CO1) (BL4)

Task 6. A model predicts class probabilities Q = (1/3, 1/3, 1/3) while the true distribution is P = (0.5, 0.25, 0.25). Compute $KL(P\|Q)$ and $KL(Q\|P)$ in bits, and analyse what the asymmetry implies when KL divergence is used as a training loss.

(3) (CO1) (BL4)

Task 7. Two binary variables A and B have the joint distribution $P(0,0) = 0.4$, $P(0,1) = 0.1$, $P(1,0) = 0.1$, $P(1,1) = 0.4$. Calculate H(A), H(B), the joint entropy H(A,B), the conditional entropy H(A|B) and the mutual information I(A;B). Analyse what the value of I(A;B) says about using B as a feature to predict A.

(4) (CO1) (BL4)

Task 8. A one-dimensional feature x follows $p(x|\omega_1) = N(2, 4)$ for class ω_1 and $p(x|\omega_2) = N(6, 4)$ for class ω_2, with priors $P(\omega_1) = 0.6$ and $P(\omega_2) = 0.4$ and zero–one loss. Derive the Bayes decision threshold x*, classify $x = 4.5$, and calculate the overall probability of error. Analyse how the threshold moves as the prior of ω_1 increases.

(4) (CO1) (BL4)

Task 9. For an observation x, the posteriors are $P(\omega_1|x) = 0.3$ (disease) and $P(\omega_2|x) = 0.7$ (healthy). Action α_1 = 'declare disease', α_2 = 'declare healthy'. The loss matrix is $\lambda(\alpha_1|\omega_1) = 0$, $\lambda(\alpha_1|\omega_2) = 8$ (unnecessary treatment), $\lambda(\alpha_2|\omega_1) = 50$ (missed disease), $\lambda(\alpha_2|\omega_2) = 0$. Evaluate the conditional risks, decide the minimum-risk action, compare it with the MAP decision and determine the posterior $P(\omega_1|x)$ above which α_1 is optimal.

(3) (CO1) (BL5)

Task 10. Polynomial regression models of increasing degree were fitted to the same dataset, giving the mean-squared errors below.

Degree	1	2	3	5	9
Training MSE	9.0	4.1	2.2	0.9	0.1

Degree	1	2	3	5	9
Validation MSE	9.5	4.6	2.9	3.6	11.8

Calculate the generalisation gap for each degree, diagnose under-fitting/over-fitting using the bias–variance trade-off, select the best model with justification, and recommend two techniques to improve the degree-9 model.

(4) (CO1) (BL5)

Task 11. A coin is tossed 10 times and shows 7 heads. The bias θ = P(head) is given a Beta(2, 2) prior. Calculate the ML estimate, the MAP estimate and the posterior mean of 0. Then evaluate how the three estimates change if the data were 70 heads in 100 tosses, and explain the role of the prior.

(4) (CO1) (BL5)

Task 12. A hospital wants an early-warning system that raises an alert when an ICU patient is likely to develop sepsis within the next 6 hours. Historical electronic health records (vital signs, lab values, outcomes) are available.

(a) Design the learning system: state the task T, performance measure P and experience E, the target function, a suitable input representation, a learning algorithm and the validation strategy. [3]

(b) On a validation cohort of 1,000 patients, 5% develop sepsis. The trained model has sensitivity 0.90 and specificity 0.80. Compute TP, FN, FP and TN. [1]

(c) Compute the positive predictive value (PPV) and negative predictive value (NPV). [1]

(d) A missed sepsis case costs ₹100 (unit cost) and a false alarm costs ₹5. Compare the total cost of (i) using the model, (ii) alerting every patient and (iii) never alerting. [2]

(e) Recommend a deployment policy and justify how the alert threshold should be set using Bayes decision theory. [1]

(8) (CO1) (BL6)

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ANSWER SCRIPT EVALUATION SHEET

ASSIGNMENT - I

Programme	B.E / B.Tech	**Course Code**	MLA02
Course Name	Fundamentals of Machine learning.	**Date**	
Duration	1 Hr 30 Min	**Max Marks**	50

Register No.: _____

TASK 1	Max Marks	Marks Awarded
Task 1	5	4
Task 2	2	2
Task 3	6	6
Task 4	4	4
Task 5	3	3
Task 6	3	3
Task 7	4	4
Task 8	4	4
Task 9	3	2
Task 10	4	4
Task 11	4	4
Task 12	8	8
TOTAL	50	48

Total in words:

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MACHINE LEARNING ASSIGNMENT-1

Task 1 - Design of Learning systems.

A retail bank plans five machine learnings systems.

a) Predict whether a loan applicant will default.

→ learning paradigm: supervised learning.

→ learning task: Binary classification.

→ Experiance (E): 10 years of labelled loan repayment records.

→ Performance (P): Accuracy, precision, recall and f1 score.

b) Discover customer behavioural segments.

→ learning paradigm: unsupervised learning.

→ Learning task: Clustering.

→ Experience (E): Behavioural data of 2 million customers without labels.

→ performance (P): silhouette score and cluster quality.

c) Estimate next month credit-card spending.

→ learning paradigm: supervised learning.

→ learning Task: Regression.

→ Experience (E): Historical customer spending records with known spending amounts.

→ performance (P): Mean Absolute Error and Root Mean squared Error (RMSE).

d) Build a fraud detector using labelled and unlabelled data.

→ learning paradigm: semi-supervised learning

→ learning task: Binary classification.

→ Experience (E): 2% labelled and 98% unlabelled tra
records
→ performance (P): Precision, recall, F1-score and PR-AUC

Task 2- Baye's Theorem

Given Disease prevalence = 2% = 0.02
sensitivity = 95% = 0.95
specificity = 90% = 0.90
false +ve rate = 1 - 0.90 = 0.10

using Baye's theorem:

$$P(D|+) = \frac{P(+|D)P(D)}{P(+|D)P(D) + P(+|\bar{D})P(\bar{D})}$$

$$= \frac{0.95 \times 0.2}{(0.95 \times 0.02) + (0.10 \times 0.98)}$$

$$= \frac{0.019}{0.019 + 0.098}$$

$$= \frac{0.019}{0.117}$$

$$= 0.1624$$

Task 3 - Joint probability Distribution

Given joint prob distribution

X	Y=0	Y=1
0	0.10	0.20
1	0.15	0.35
2	0.20	0.10

Amila Brotheer

Marginal probabilities of X:

$P(X=0) = 0.10+0.20 = 0.30$

$P(X=1) = 0.15+0.25 = 0.40$

$P(X=2) = 0.20+0.10 = 0.30$

Marginal probabilities of Y:

$P(Y=0) = 0.10+0.15+0.20 = 0.45$

$P(Y=1) = 0.20+0.25+0.10 = 0.55$

b) calculate E[X], var[X] and E[Y]

$E[X] = \Sigma x p(x)$

$= 0(0.30)+1(0.40)+2(0.30)$

$= 1.00$

$E[X^2] = 0^2(0.30)+1^2(0.40)+2^2(0.30)$

$= 1.60$

Var[X]

$var[X] = E[X^2] - (E(X))^2$

$= 1.60 - 1^2 = 0.60$

$= 0.60$

E[Y]

$= 0.(0.45)+1(0.55)$

$= 0.55$

c) conditional prob

$P(X=2|Y=1) = \frac{P(X=2, Y=1)}{P(Y=1)} = \frac{0.10}{0.55} = 0.1818$

d) covariance and correlation coeff

$E[XY] = 0+0.25+0.20 = 0.45$

covariance:

$cov(X,Y) = E(X,Y) - E(X)E(Y)$

$= 0.45 - 1(0.55)$

$= -0.10$

variance of Y:

$Var(Y) = 055(0.45) -$

$= 0.2475.$

correlation coeff

$\gamma = \frac{cov(X,Y)}{\sqrt{var(X)\, var(Y)}}$

$= \frac{-0.10}{\sqrt{0.60 \times 0.2475}}$

$= -0.2595$

e) check independence!

$P(x,y) = P(x)\,P(y)$

$P(x=0, y=0) = 0.10$

But $P(x=0)\,P(y=0) = 0.30 \times 0.45 = 0.135$

$\therefore$ 0.10 is not equal to 0.135, x and y are not independent.

Task 4 – Normal Distribution and Max likelihood Estimation

Given: $x \sim N(50, 16)$ Mean = 50 & Variance = 16

SD $\Rightarrow \sigma = \sqrt{16} = 4$

a) cal $P(x > 58)$

using z-score:

$$z = \frac{x-\mu}{\sigma} = \frac{58-50}{4}$$

$z = 2$

$P(x > 58) = 1 - \phi(2)$

$= 1 - 0.9772 = 0.0228$

cal $P(46 < x < 54)$:

$$z_1 = \frac{46-50}{4} = -1$$

$$z_2 = \frac{54-50}{4} = 1$$

$P(-1 < z < 1) = 0.8413 - 0.1587$

$= 0.6826$

b) Max likelihood Estimates

Given readings: 12, 15, 11, 14, 13

Mean: $\mu_{ML} = \frac{65}{5} = 13$

ML variance: $\sigma^2_{ML} = \frac{\Sigma (x_i - \mu)^2}{n}$

$= \frac{1+4+4+1+0}{5}$

$= \frac{10}{5}$

$= 2$

unbaised variance

$s^2 = \frac{\Sigma (x_i - \bar{x})^2}{n-1}$

$= \frac{10}{4}$

$= 2.5$

5 - Information theory and Entropy

Given probabilities:

$$P(1/2, 1/4, 1/8, 1/8)$$

a) cal entropy

$$H(X) = -\sum P(x) \log_2 P(x)$$

$$= -\left[\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{8}\log_2\frac{1}{8} + \frac{1}{8}\log_2\frac{1}{8}\right]$$

$$= 0.5 + 0.5 + 0.375 + 0.375$$

$$H(X) = 1.75 \text{ bits}$$

b) Compare with uniform distribution

$$H(X) = \log_2 4 = 2 \text{ bits}$$

$$R = 2 - 1.75$$

$$= 0.25 \text{ bits}$$

% reductancy:

$$= \frac{0.25}{2} \times 100$$

$$= 12.5\%$$

c) Avg code length

Given code: 111

code lengths = 1, 2, 3, 3

$$L = \sum P(x_i) l_i$$

$$= \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{8}(3)$$

$$= 0.5 + 0.5 + 0.375 + 0.375$$

$$L = 1.75 \text{ bits.}$$

Task 6 - KL Divergence

a) cal KL(P||Q)

Formula: $D_{KL}(P||Q) = \sum P(x) \log_2 \frac{P(x)}{Q(x)}$

$$= 0.5 \log_2(1.5) + 0.25 \log_2(0.75) + 0.25 \log_2(0.75)$$

$$= 0.08496 \text{ bits}$$

b) cal KL(Q||P) => $D_{KL}(Q||P) = \sum Q(x) \log_2 \frac{Q(x)}{P(x)}$

$$= \frac{1}{3}\log_2\frac{2}{3} + \frac{1}{3}\log_2\frac{4}{3} + \frac{1}{3}\log_2\frac{4}{3} = 0.08170 \text{ bits}$$

c) Explain the asymmetry

KL divergence is not symmetric.

$D_{KL}(P||Q) \neq D_{KL}(Q||P)$

The 2 values are diff because reversing the true and predicted distributions changes the calculation

Task 7 - Entropy and Mutual information

Given:

A	B	Prob
0	0	0.4
0	1	0.1
1	0	0.1
1	1	0.4

Marginal prob:

$P(A=0) = 0.5, \quad P(A=1) = 0.5$

$P(B=0) = 0.5, \quad P(B=1) = 0.5$

a) Cal $H(A)$ & $H(B)$

$H(A) = -0.5\log_2(0.5) - 0.5\log_2(0.5)$

$H(A) = 1$ bit

Similarly $H(B) = 1$ bit

b) Joint entropy $H(A,B)$

$H(A,B) = -\Sigma P(A,B)\log_2 P(A,B)$

$= -[2(0.4\log_2 0.4) + 2(0.1\log_2 0.1)]$

$H(A,B) = 1.7219$ bits.

c) conditional entropy $H(A|B)$

$H(A|B) = H(A,B) - H(B)$

$= 1.7219 - 1$

$= 0.7219$ bits

d) Mutual information

$I(A;B) = H(A) - H(A|B)$

$= 1 - 0.7219$

$= 0.2781$ bits

Bayes Decision Theory

Given: $P(x/w_1) = N(2,4)$ Priors: $P(w_1) = 0.6$

$P(x/w_2) = N(6,4)$ $P(w_2) = 0.4$

a) Derive the Bayes decision threshold:

For 0-1 loss choose the class with the higher posterior prob.

$$P(w_1)P(x/w_1) = P(w_2)P(x/w_2)$$

Taking logarithms and simplifying:

$$\ln\frac{0.6}{0.4} - \frac{(x-2)^2}{8} + \frac{(x-6)^2}{8} = 0$$

$$\ln(1.5) - x + 4 = 0$$

$$\therefore x^* = 4 + \ln(1.5)$$

$$\boxed{x^* = 4.4055}$$

b) classify $x = 4.5$

$\because 4.5 > 4.4055$

observation $\in$ class:

$\boxed{w_2}$

c) cal over prob of error

$$P(\text{error}) = P(w_1)P(x > x^*/w_1) + P(w_2)P(x < x^*/w_2)$$

$$z_1 = \frac{4.4055 - 2}{2} = 1.203$$

$$z_2 = \frac{4.4055 - 6}{2} = -0.797$$

$$P(\text{error}) = 0.6(0.1145) + 0.4(0.2126)$$

$$= 0.0687 + 0.0850$$

$$P(\text{error}) \approx 0.1538.$$

d) effect of ↑ prior prob

If the prior prob of class w_1 increase, the decision threshold moves towards the right. This means more observations will be classified as class w_1.

Task-9 - min Risk Decision

Given : $P(w_1|x) = 0.3$ $P(w_2|x) = 0.7$

loss matrix:

Action	Disease	Healthy
declare disease (α_1)	0	8
declare healthy (α_2)	50	0

a) Cal conditional risk

$R(\alpha_1|x) = 0(0.3) + 8(0.7)$

$= 5.6$

$R(\alpha_2|x) = 50(0.3) + 0(0.7)$

$= 15$

b) min risk action

$\because 5.6 < 15$

α_1 = declare disease

c) compare with MAP decision

MAP chooses the class with highest posterior prob

$\therefore$ healthy prob is 0.7 MAP chooses healthy

d) posterior threshold

$8(1-P) < 50P$

$8 < 58P$

$P > \frac{8}{58}$

$P > 0.2759$

Task 10 - Bias variance Trade off

Given training and validation MSE

a) Cal generalisation gap

gap = validation MSE - Training

Degree	Calculation	gap
1	9.5 - 9.0	0.5
2	4.6 - 4.1	0.5
3	2.9 - 2.2	0.7
5	3.6 - 0.9	2.7
9	11.8 - 0.1	11.7

...ose under-fitting and over fitting

degree 1 : under-fitting tracing and validation errors are high.

→ Degree 2 : good balance blw bias & variance

→ Degree 3 : good performance with a small generalisation gap

→ Degree 5 : OF beging Lower validation error increase.

→ Degree 9 : Severe OF Cause TE is very low but VE is very high.

c) select the best Model

Degree 2 has lowest validation MSE

Actually degree 3 has the lowest validation MSE.

Validation MSE = 2.9

∴ Degree 3 is most best Model

d) Two techniques to improve the degree 9 Model

1. Regularization : Apply L1 or L2 regularisation to reduce Model complexity & prevent overfitting.

2. Reduce polynomical degree : use a lower-degree polynomical or feature selection to make the model simpler

Task II - Bayesian Estimation

Given no of coin tosses = 10

no of heads = 7

tails = 3

Prior distribution = Beta (2,2)

a) Max likelihood Estimate

$$\hat{\theta}_{ML} = \frac{H}{N} = \frac{7}{10} = 0.7$$

$$\hat{\theta}_{MAP} = \frac{\alpha' - 1}{\alpha' + \beta' - 2}$$

$$= \frac{9-1}{9+5-2}$$

$$= \frac{8}{12}$$

$$= 0.6667$$

b) Max A posteriority Estimate

Beta $(\alpha + H, \beta + T)$ = Beta $(2+7, 2+3)$

Beta (9,5)

c) Posterior Mean

$$E[\theta|D] = \frac{\alpha'}{\alpha + \beta'}$$

$$= \frac{9}{14}$$

$$= 0.6429$$

d) if 70 heads occur in 100 tosses

Estimate	10 tosses	100 tosses
ML	0.9	0.7
MAP	0.667	0.6961
posterior Mean	0.6429	0.6923

Task 12 - Hospital sepsis Early Early-warning system

a) Design the learning system

1. Task (T): Predicts whether an ICU patient will develop sepsis within the next 6 hours. it is a binary classification problem.
2. Performance measure (P): sensitivity specificity precision, recall F1-score and ROC-AUC.
3. Experience (E): Historical electronic health records containing vital signs, laboratory values and patient outcomes.

b) cal TP, FN, FP and TN

Given Total patients = 1000

sepsis prevalence = 5%

sensitivity = 0.90

specificity = 0.80

+ve patients: $1000 \times 0.05 = 50$

-ve patients: $1000 - 50 = 950$

True +ve: $TP = 50 \times 0.90 = 45$

True -ve: $TN = 950 \times 0.80 = 760$

False -ve: $FN = 50 - 45 = 5$

False +ve: $FP = 950 - 760 = 190$

Final confusion Matrix

	Actual sepsis	Actual Healthy.
Predicted sepsis	TP = 45	FP = 190
Predicted Healthy	FN = 5	TN = 76

PPV and NPV

$$PPV = \frac{TP}{TP+FP} = \frac{45}{45+190} = 0.1915$$

$$NPV = \frac{TN}{TN+FN} = \frac{760}{760+5} = 0.9935$$

d) cal total cost

given: cost of missed sepsis case = ₹100

cost of false alarm = ₹5

case 1: using Model

Missed cases = 5

False alarms = 190

cost = (5×100) + (190×5)

= 500 + 950

= 1450

case 2: Alerting every patient

cost = 950×5

Total = 4750

case 3: Never alerting

cost = 50×100

Total = 5000

e) Recommend a deployment policy

let p be prob of sepsis.

$5(1-P) < 100P$

$5 < 105P$

$P > 0.0476$